

Protocol Stack

Module 05

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my dear students hello I welcome you all in our series of lectures on cyber security today we are going to study about protocol stack after going through this lesson you will be able to understand the needs of protocols and standards also you will be able to identify various protocols at different layers of protocol stack so let us begin with it the protocol stack is an implementation of a computer networking protocol suit the terms are often used interchangeably the suit is the definition of the protocols and the stack is the software implementation of them a protocol stack is a complete set of network protocol layers that work together to provide networking capabilities it is called aack because

it is typically designed as a hierarchy of layers each supporting the one above it and using those below it individual protocols within a suit are often designed with a single purpose in mind this modelization makes design and EV valuation easier because each protocol module usually communicates with two others they're commonly imagined as layers in a stack of protocols the lowest protocol always deals with low level physical interaction of the hardware every higher layer adds more features user applications usually deal only with the topmost layers in Practical implementation 2 minutes, 1 second protocol Stacks are often divided into three major sections media transport and applications a particular operating

system or platform will often have two well-defined software interfaces one between the media and transport layers and one between the transport layers and applications the media to transport interface defines how transport protocol software makes use of particular media and Hardware types for example this interface level would Define how TCP IP transport software would talk to the ethernet Hardware the application to transport interface defines how application programs make use of the transport layers for example this interface level would Define how a web browser program would talk to the TCP IP transport software next is OSI model 3 minutes, 1 second Osa stands for Open System interconnection model it is a conceptual model that characterizes and standardizes the communication functions

of a telecommunication or Computing system without regard to their underlying internal structure and Technology its goal is the interoperability of diverse communication systems with the standard protocols the model partitions a communication system into abstraction

layers the original version of the model defines seven layers the model is a product of the open systems interconnection project at the International Organization for standardization maintained by identification ISO 7498 the Osa model has seven layers the principles that were applied to arrive at the seven layers can be briefly summarized as number one a layer should be created where a different abstraction is needed number two each layer should perform a well- defined

function number three the function of each layer should be chosen with an eye toward defining internationally standardized protocols and number four the layer boundaries should be chosen to minimize the information flow across the interfaces number five the number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy figure shows the OSI reference model see this next figure the figure shows the exchange of data using USA model D7 means the data unit at L 7 D6 means the data unit at layer six and so on the process starts at layer 7 called the application layer and then moves from layer to layer in descending

sequential order at each layer a header or possibly a trailer can be added to the data unit commonly the trailer is added only at Layer Two when the formatted data unit passes through the physical layer it is changed into an electromagnetic signal and transported along a physical link upon reaching its destination the signal passes into layer one and is transformed back into digital form the data units then move back up through the OSI layers as each block of data reaches the next higher layer the headers and trailers attached to it at the corresponding sending layer are removed and actions appropriate to that layer are taken by the time it reaches L 7 the 6 minutes, 1 second message is again in a form appropriate to the application and is made available to the

recipient next is layers in the Osa model the various layers of Osa model are number one physical layer in the seven layer USA model of computer networking the physical layer or layer one is the first and the lowest layer the physical layer coordinates the functions required to carry a bitstream over a physical medium it deals with a mechanical and electrical specifications of the interface and transmission medium it also defines the procedures and functions that physical devices and interfaces have to perform for transmission to occur now what are the services provided by physical layer in digital Communications symbol rate also known as board rate and modulation rate is the number of symbol changes waveform changes or signaling

events across the transmission medium per time unit using a digitally modulated signal or a line code the symbol rate is measured in board or symbols per second in the case of a line code the symbol rate is the pulse rate in pulses per second number two providing a standardized interface to physical transmission media including mechanical specifications of electrical connectors and cables for example maximum cable length providing electrical

specifications of transmission line signal level and impedance providing radio interface including electromagnetic spectrum frequency allocation and specification of signal strength analog bandwidth extra and providing specifications for IR over optical fiber or a wireless IR communication link

8 minutes, 1 second number three modulation in electronics and Telecommunications modulation is the process of varying one or more properties of a periodic waveform called the carrier signal with a modulating signal that typically contains information to be transmitted a modulator is a device that performs modulation a demodulator is a device that performs demodulation the inverse of modulation a botom can perform both operations number four line coding line coding consists of representing the digital signal to be transported by a waveform that is optimally tuned for the specific properties of the physical Channel and of the receiving equipment the pattern of voltage current or Photon used to represent the digital data on transmission link is called the

line encoding the common types of line encoding are unipolar polar bipolar and Manchester encoding after line coding the signal is put through a physical Channel either a transmission medium or a data storage medium number five bit synchronization the sender and receiver not only must use the same bit rate but also must be synchronized at the bit level in other words the and the receiver clocks must be synchronized number six line configuration the physical layer is concerned with the connection of devices to the media in a point to-point configuration two devices are connected through a dedicated Link in a multipoint configuration a link is shared among several devices number seven physical

topology the physical topology defines how devices are connected to make a network and number eight is transmission mode transmission mode defines the direction of transmission between two devices Simplex half duplex or full duplex next is data link layer the data link cayer or Layer Two is the second layer of the seven layer USA model of computer networking this layer is the protocol layer that trans transfers data between adjacent Network nodes in a wide area network or between nodes on the same local area network segment the data link clear provides the functional and procedural means to transfer data between Network entities and might provide the means to detect and possibly correct errors that may occur in the physical layer now what are the services provided by data link

CER number one framing a frame is the protocol data unit at the data link cayer frames are the result of the final layer of encapsulation before the data is transmitted over the physical layer a frame is a unit of transmission in a link layer protocol and consist of a link clear header followed by a packet number two physical addressing if frames are to be distributed to different systems on the network the data link clear adds a header to the frame to define the sender or receiver of the frame if the frame is intended for a system outside the senders Network the receiver address is the address of the device that connects the network to the next one number three flow control flow control is the process of managing the rate of data

transmission between two nodes to prevent a fast sender from

overwhelming a slow receiver it provides a mechanism for the receiver to control the transmission speed so that the receiving node is not overwhelmed with data from the transmitting node number four error control it allows the receiver to inform the sender if a frame is lost or damaged during transmission and coordinates the retransmission of those frames by the sender error control in the data link layer is based on automatic repeat request whenever an error is detected specified frames are retransmitted number five logical link control logical link control refers to the functions required for the establishment and control of logical links between local devices on a network as mentioned this is usually considered a data link sublayer it provides services to the network layer above it

and hides the rest of the details of the data link layer to allow different Technologies to work seamlessly with the higher layers most local area networking Technologies use the IEEE 802.2 LLC protocol and number six is Media Access Control this refers to the procedures used by devices to control access to the network medium since many networks use a shared medium such as a single network cable or a series of cables that are electrically connected into single virtual medium it is necessary to have rules for managing the medium to avoid conflicts for example ethernet uses the CSMA/CD method of media Access Control while token ring uses token passing next is Network layer in the seven layer OSI model of computer

networking the network layer is layer three the network layer is responsible for packet forwarding including routing through intermediate routers since it knows the address of neighboring Network nodes and it also manages quality of service the services provided by Network layer are number one logical addressing every device that communicates over a network has associated with it a logical address sometimes called a layer three address for example on the internet the Internet Protocol is the network layer protocol and every machine has an IP address number two routing moving data across a series of interconnected networks is probably the defining function of the network layer it is the job of the devices and software routines that function at the network layer to handle incoming Packets

from various sources and determine their final destination number three datagram encapsulation the network layer normally encapsulates messages received from higher layers by placing them into datagrams also called Packets with a network layer header number four fragmentation and reassembly the network layer must send messages down to the data link layer for transmission some data link layer technologies have limits on the length of any message that can be sent if the packet that the network layer wants to send is too large the network layer must split the packet up send each piece to the data link layer and then have pieces reassembled once they arrive at the network layer on the destination machine number five error handling and

Diagnostics special Protocols are used at the network layer to allow devices that are logically connected or that are trying to route traffic to exchange information about the status of host on the network or their devices themselves various protocols at Network Layer are ICMP internet control message protocol IGMP internet group management protocol IPsec Internet Protocol security IP version 4 or IP version 6 that is Internet Protocol next is transport layer the transport layer is called the host to host transport layer in the TCP/IP model the data communicated by the transport layer is encapsulated in a transport layer PDU and sent in a network layer the network layer nodes transfer the transport PDU intact without decoding or

modifying the content of the PDU in this way only the peer transport entities actually communicate using the PDUs of the transport protocol it provides services such as connection oriented data stream reliability flow control and multiplexing let us discuss them one by one number one connection oriented communication it is normally easier for an application to interpret a connection as a data stream rather than having to deal with the underlying connectionless models such as the datagram model of the user datagram protocol and of the Internet Protocol number two same order delivery the network layer doesn't generally guarantee that packet of data will arrive in the same order that they were sent but often this is a desirable feature this is usually done through the use of segment numbering with the

receiver passing them to the application in order number three reliability Packets may be lost during the transport due to network congestion and errors by means of an error detection code such as checksum the transport protocol may check that the data is not corrupted and verify Cryptographic by sending an ACK or NACK message to the sender automatic repeat request schemes may be used to retransmit lost or corrupted data number four flow control the rate of data transmission between two nodes must sometimes be managed to prevent a fast sender from transmitting more data than can be stored by the receiving data buffer causing a buffer overrun this can also be used to improve efficiency by reducing buffer underrun

number five congestion avoidance congestion control can control traffic entry into a telecommunications Network so as to avoid congestive Collapse by attempting to avoid oversubscription of any of the processing all link capabilities of the intermediate nodes and networks and taking resource reducing steps such as reducing the rate of sending packets for example automatic repeat request may keep the network in a congested state for 19 minutes, 1 second this situation can be avoided by adding congestion avoidance to the flow control including slow start this keeps the bandwidth consumption at a low level in the beginning of the transmission or after packet retransmission and number six is multiplexing ports can provide multiple end points on a single node for example the name on a postal address is a kind

of multiplexing and distinguish is between different recipients of the same location Computer Applications will each listen for information on their own ports which enables the use of more than one network service at the same time it is the part of the transport layer in

the tcpip model but of the session layer in the Osa model let us now learn about two most important protocols at transport layer number one transmission control protocol TCP is a connection oriented protocol which means a connection is established and maintained until the application programs at each end have finished exchanging messages it determines how to break application data into packets that networks can deliver sends packet to and accepts packet from the network CER

manages flow control and because it is meant to provide error free data transmission handles retransmission of dropped or garbled packets as well as acknowledgement of all packets that arrive and second is user datagram protocol the user datagram protocol is a transport layer protocol defined for use with the IP network layer protocol it provides a best effort datagram service to an end system the service provided by UDP is an unreliable service that provides no guarantees for delivery and no protection for duplication the Simplicity of UDP reduces the overhead from using the protocol and the services 21 minutes, 1 second may be adequate in many cases UDP provides a minimal unreliable best effort message passing transport to

applications and upper layer protocols compared to other transport protocols UDP and its UDP light variant are unique in that they do not establish endtoend connections between communicating and systems UDP communication consequently does not incur connection establishment and tear down overheads and there is minimal Associated end System state because of these characteristics UDP can offer a very efficient communication transport to some applications but has no inherent congestion control or liability on many platforms applications can send UDP datagrams at the line rate of the link interface which is often much greater than the available path capacity and doing so would contribute to congestion

along the path applications therefore need to be designed responsibly next is session layer the session layer provides the mechanism for opening closing and managing a session between end user application processes that is a semi-permanent dialogue communication sessions consist of requests and responses that occur between applications session layer services are commonly used in application environments that make use of remote procedure calls now the services provided by session layer are number one authentication number two authorization and number three session restoration the session layer of the USA model is responsible for session checkpointing and Recovery it allows information of different streams perhaps originating from different sources to be properly

combined or synchronized an example usage of the session layer is session beans which are only active as long as the session is active and are deleted when the session is disconnected developers can use them to store information about the user during a web session an example application is web conferencing in which the streams of audio and video must be synchronous to avoid so-called lip sync problems flow control ensures that the person displayed on screen is the current Speaker another application is in live TV programs where streams of audio and video need to be seamlessly merged and transitioned from one to the other to avoid silent air

time or excessive overlap next is presentation layer the primary goal of this layer is to take care of the syntax and semantics of the

information exchanged between two communication systems presentation layer takes care that the data is sent in such a way that the receiver will understand the information and will be able to use the data language syntax can be different of the two communicating systems under this condition presentation layer plays a role translator the various functions of presentation layer are number one translation before being transmitted information in the form of characters and numbers should be changed to bit frames the presentation layer is responsible for interoperability between encoding methods as different computers use different encoding methods it translates data between the formats the network requires and then the data

according to the computer number two encryption the presentation layer carries out encryption at the transmitter and decryption at the receiver and number three is compression it carries out data compression to reduce the bandwidth of the data to be transmitted the primary role of data compression is to reduce the number of bits to be transmitted it is important in transmitting multimedia such as audio video text Etc last but not the least is application layer it is the topmost layer of OSI model manipulation of data or information in various ways is done in this layer which enables user or softwares to get access to the network some Services provided by this layer includes email transferring of files Distributing the results to the user directory Services network resources at

stra various functions provided by application layer are number one mail services this layer provides the basis for email forwarding and storage number two Network virtual terminal it allows a user to log onto a remote host the application creates software emulation of a terminal at the remote host users computer talks to the software terminal which in turn talks to the host and vice versa then the remote host believes it is communicating with one of its own Terminals and allows users to log on number three directory services this layer provides access for Global Information about various services and 26 minutes, 1 second number four is file transfer access and management it is a standard mechanism to access files and manage it users can

access files in a remote computer and manage it they can also retrieve files from a remote computer next is the TCP IP reference model the TCP IP reference model is the network model used in the current Internet architecture the reference model was named after two of its main protocols TCP transmission control protocol and IP that is Internet Protocol the developers choose to build a pack switch network based on a connectionless inter Network layer the figure shows various layers of TCP IP reference model let us now briefly discuss about the various layers of TCP IP model number one the host to network layer that is physical layer the TCP IP reference model does not specify in any great detail the operation of this layer except that the host has to connect to the network using some protocol so it can send IP packets

over it number two the network layer the job of the network layer is to inject Packets into any network and have them travel independently to the destination the layer defines IP for its official packet format and protocol packet routing is a major job of this protocol number three the transport layer the transport layer is the interface between the application layer and the complex Hardware of the Network it is designed to allow peer entities of the source and destination host to carry on conversations data may be user data or Control Data two modes are available full duplex and half duplex in full duplex operation both sides can transmit and receive data simultaneously whereas in half duplex a site can only send or receive at one time any program running in the

application layer has the ability to send a message using TCP or UDP which are the two protocols defined for the transport layer the application can communicate with the TCP or the UDP service whichever it requires both the TCP and UDP communicate with the Internet Protocol in the internet layer in all cases communication is a two-way process the applications can read and write to the transport layer number four the application layer the original TCP IP specification described a number of different applications that fit into the top layer of the protocol stack these applications include telnet FTP SMTP and DNS telnet is a program that supports the telnet protocol over TCP telnet is a general two-way

communication protocol that can be used to connect to another host and run applications on that host remotely FTP that is file transfer protocol is a protocol that was originally designed to promote the sharing of files among computer users it Shields the user from the variations of file storage on different architectures and allow for a reliable and efficient transfer of data SMTP that is simple mail transfer protocol is the protocol used to Transport electronic mail from one computer to another through a series of other computers along the road DNS domain name system resolves the numerical address of a network node into its textual name or vice versa it would translate www.yahoo.com to 204.

71177 71 to allow the routing protocols to find the host that the packet is designed for now what are the merits of tcpip number one it operates independently number two it is scalable number three client server architecture number four Supports a number of routing protocols and number five can be used to establish a connection between two computers next is Demerits of tcpip number one in this the transport layer does not guarantee delivery of packets number two the model cannot be used in any other application number three replacing protocols is not easy and number four it has not clearly separated its services interfaces and protocols so with this our today's lecture is over concluding this today we learned about the need of protocols and standards also we learned about about

various protocols at different layers of protocol stack hope you understood today's lecture in the next lecture we'll learn about few more concepts of networks and cyber security till then goodbye thank you have a nice day